

California 2021 Benchmark Dataset: Construction (Appendix)

This appendix describes how we built the **California 2021** wildfire ignition dataset and the **operational domain mask** used with the Pyrologix static risk map in benchmarks such as `benchmark_2021_greedy_kernel/benchmark_2021_greedy_kernel.md`. Figures are generated by:

```
conda run -n wf python code/dataset_creation/nature_dataset_creation/generate_paper_2021_dataset_explain.py
```

(Plots use `plot_pyrologix_valid_region` and `plot_pyrologix_fire_categories` in `code/displays.py`.)
Auto-updated cell and fire counts are in `counts.md`.

1. Risk map and study grid

Pyrologix ignition probability, resampled to the same **1 km** grid as the USGS WFPI product (**1309** $\times$ **805** cells, Lambert Azimuthal Equal-Area, California bounding box with 50 km buffer), is stored in `California2021Dataset/static_risk_pyrologix.npy`. The model was trained on **2006–2020** data, so it is **out-of-sample for 2021** ignitions (no training leakage on evaluation fires). Values are rescaled to **0–255** for compatibility with placement code.

2. Mask construction (cumulative steps)

The **final mask** copied into `California2021Dataset/mask.npy` is `California2020Dataset/mask_union_burnable_no_snow`; it is **independent of 2021** and defines where ignitions and routing are valid. We visualize the same logic **cumulatively** on a Pyrologix background (invalid cells are **white** in the figures). **Figures use a linear scale 0–1** (stored values 0–255 divided by 255); the color scale is inset at the upper right of each map. **Dataset ignitions** (black markers) are overlaid on every map, including the four mask steps below. **Fire-category legends** match the all-categories figure: **urban** (teal triangles), **Unburnable** (grey disks, once the WFPI rule applies), and **in dataset** (black disks). Ignitions outside the California polygon are **not** drawn on the maps (they remain in the §3 count table). For the first two mask panels (before the WFPI union-burnable constraint), the legend **omits** the Unburnable row; those ignitions are still plotted as ordinary black dots (see figure captions).

Step	Description
(1)	California land: rasterized US Census tract boundary (state polygon), 50 km buffer on the WFPI crop.
(2)	Exclude urban: cells overlapping US Census Urban Areas 2020 (<code>t1_2025_us_uac20</code>).
(3)	Exclude permanently non-burnable (WFPI): using all 366 daily WFPI Day 1 maps for 2020 , a cell is kept if on at least one day it has value < 249 or = 250 (snow); cells that are always in 249 or 251–255 are removed (“union burnable”, snow not treated as invalid).
(4)	Largest connected component: small islands and fragments are removed; the main California landmass remains.

Valid cells after each step

Step	Valid cells
(1) California boundary	426,469
(2) Excluding urban	397,815

Step	Valid cells
(3) Excluding always WFPI-invalid	266,399
(4) Largest connected component	249,255

3. Fire records: USFS ignitions (2021)

Source: USFS ignition-point extract `USFS_ignition_points.csv` (wildfire records with discovery location and time).

Stage 1 — tabular and geographic filters

- Year **2021**, California identifiers (UNIQFIREID prefix 2021-CA), **wildfire** only (FIRETYPECATEGORY == 'WF').
- Valid **discovery datetime**, **latitude/longitude**.
- Point inside **California** state polygon (Census tracts, dissolved).
- **Not** inside an **urban** polygon (same UAC layer as the mask).

Stage 2 — grid and mask

- Project each point to WFPI grid indices (**row**, **col**).
- Discard if out of bounds or if `mask[row, col] != 1`.

2021-specific exclusion

- For **27** calendar days, no native **2021 WFPI Day 2** archive zip is available. Ignitions whose **discovery date** falls on one of those days are **excluded** so every retained scenario can be paired with real or nearest-neighbour WFPI files (see `create_california_2021_dataset.py`). They appear as a separate category on the overview figure.

Fire counts

Category	Count
Raw CA wildfires (CSV filters)	1,086
Outside CA boundary	13
Urban	25
Non-urban, off grid or mask	63
Excluded — missing 2021 WFPI zip date	53
In dataset (filter pipeline)	932
Scenario files <code>*.npz</code> on disk	931

The spatial filter yields **932** ignitions; the repository currently contains **931** per-scenario files under `California2021Dataset/scenarii/` (see `dataset_summary.json`). Benchmark code draws only from scenarios that have `offset_ / date_ / time_` entries in `config_california_2021.json`.

4. Fire figures (Pyrologix, final mask)

All categories — ignitions from the stage-1 cohort shown by filter outcome: urban (teal), **Unburnable** (grey), and **in dataset** (outside-CA points are omitted from the figure). (Fires excluded only because of missing 2021 WFPI zip dates are omitted from this map; their count is in the table in §3.)

Dataset only — ignitions that pass all filters (**932** in the pipeline; coordinates are those used for mask checks).

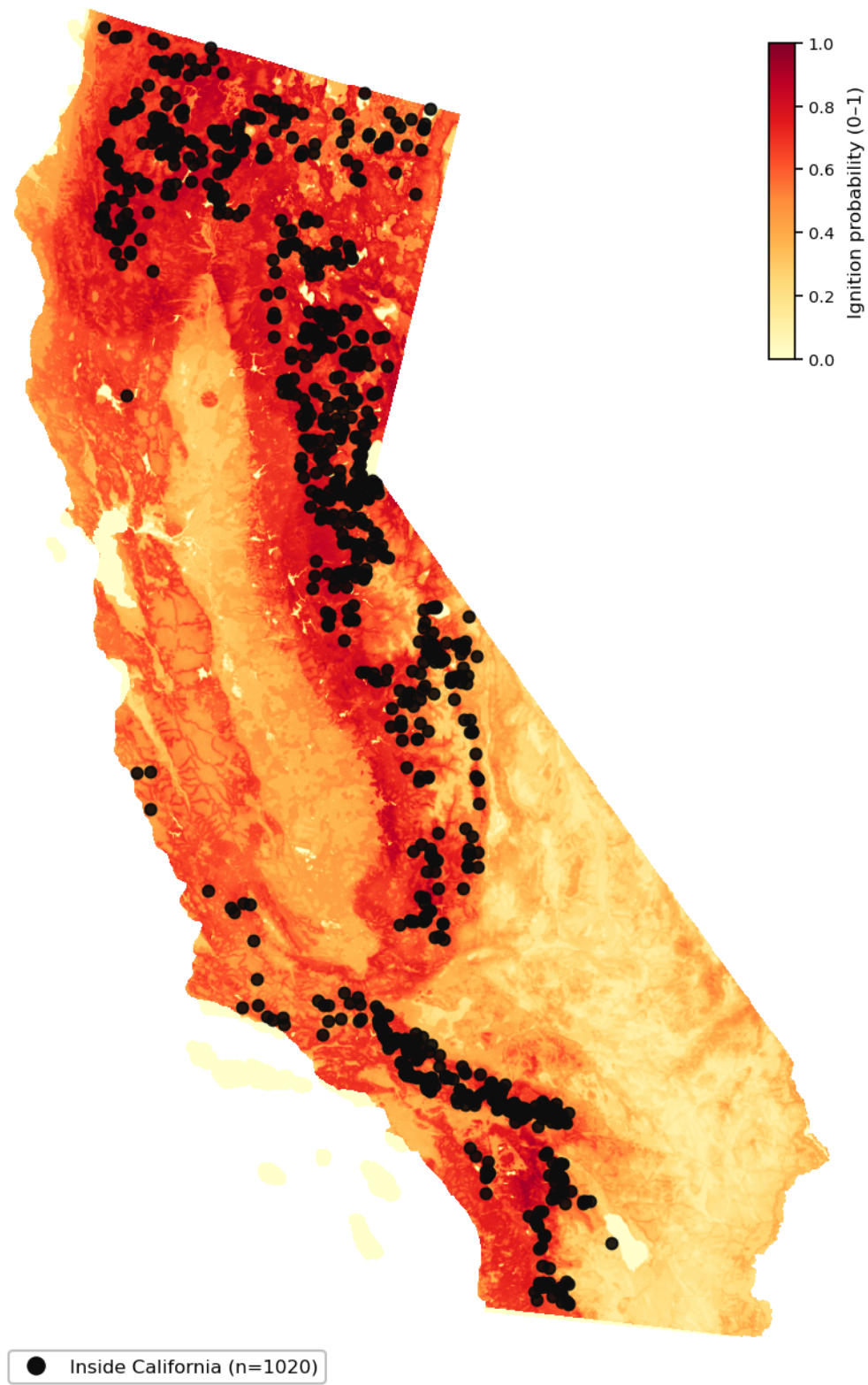


Figure 1: (1) California boundary

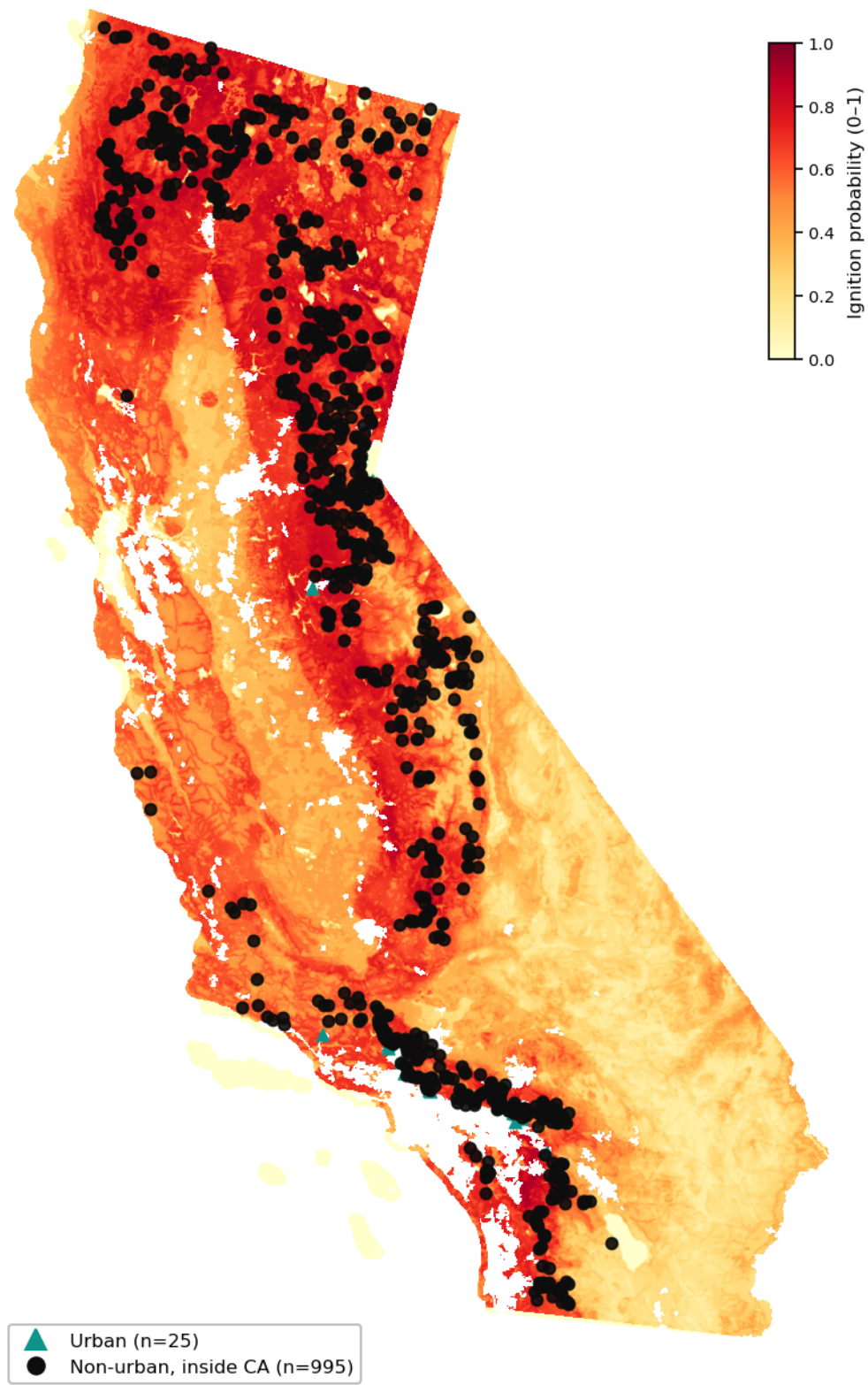


Figure 2: (2) Exclude urban

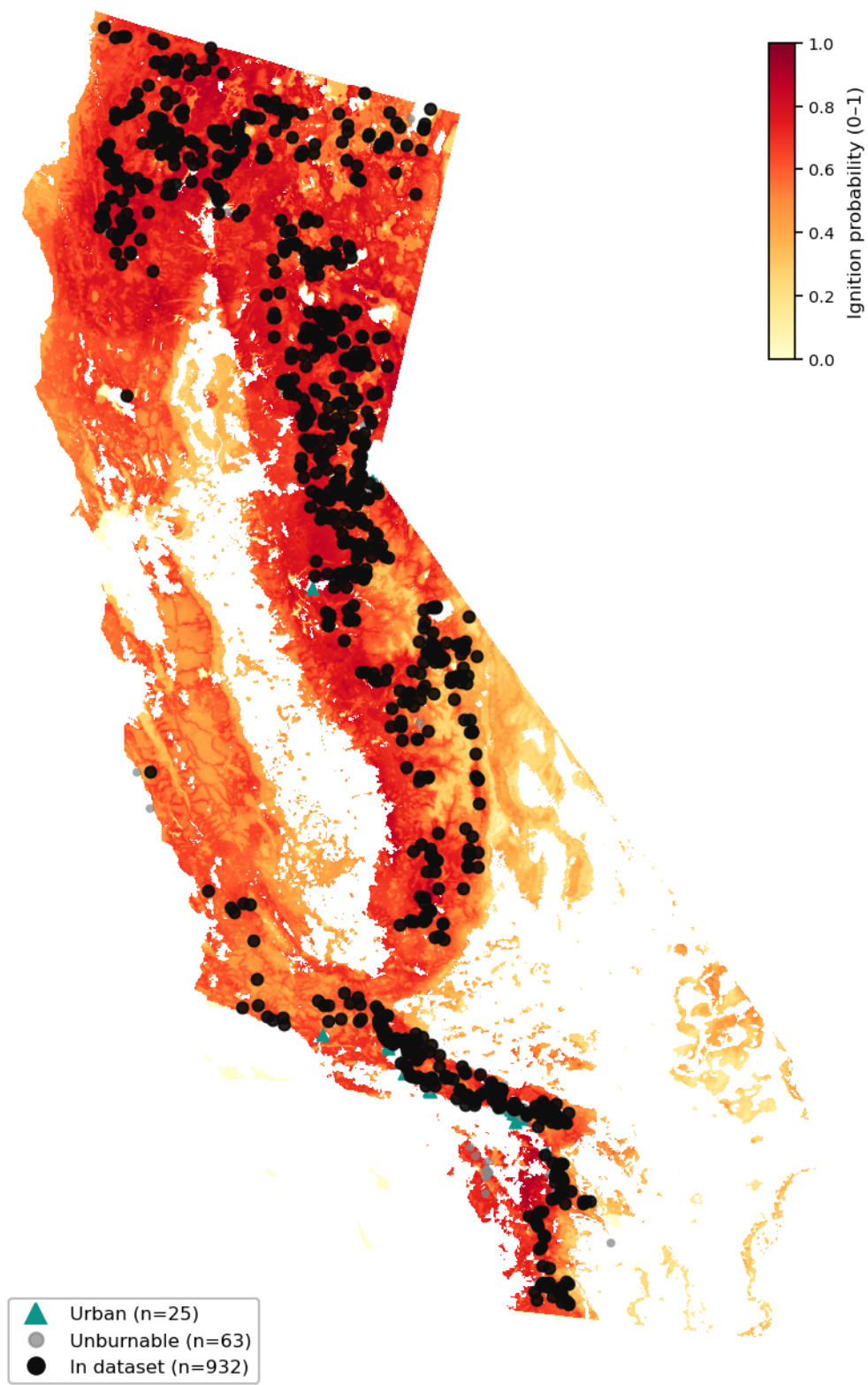


Figure 3: (3) Exclude always WFPI-invalid

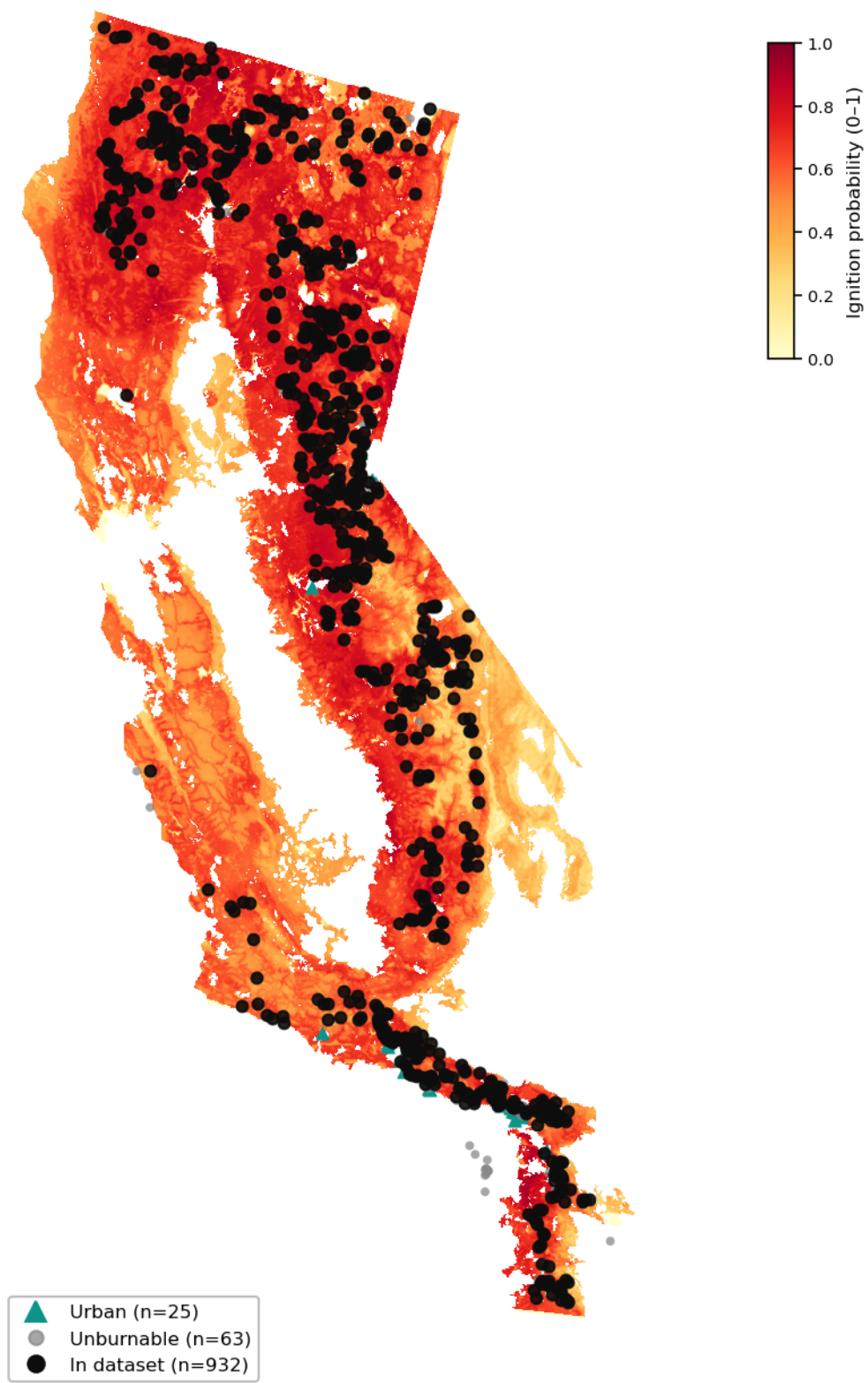


Figure 4: (4) Final mask — largest CC

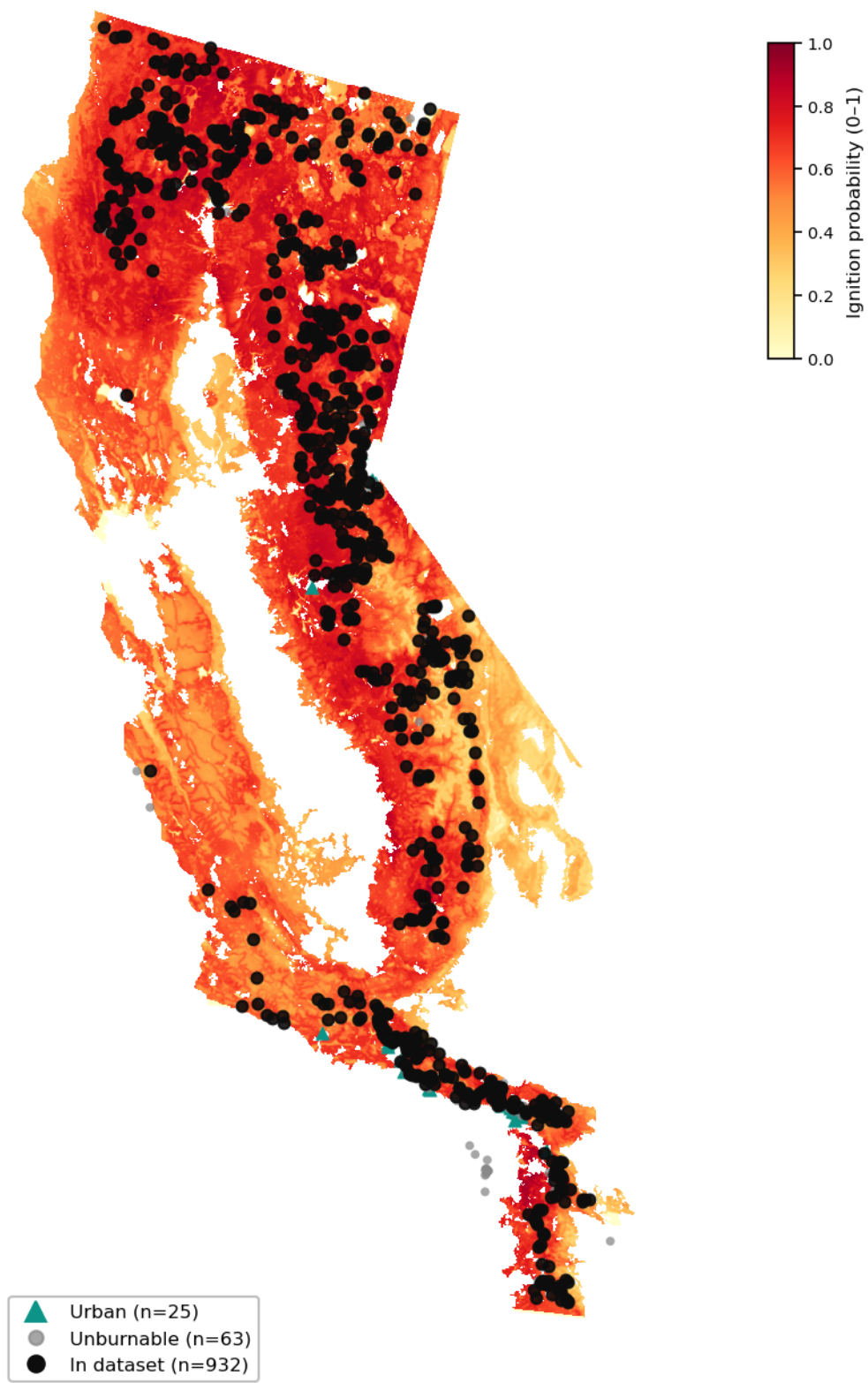


Figure 5: All fire categories

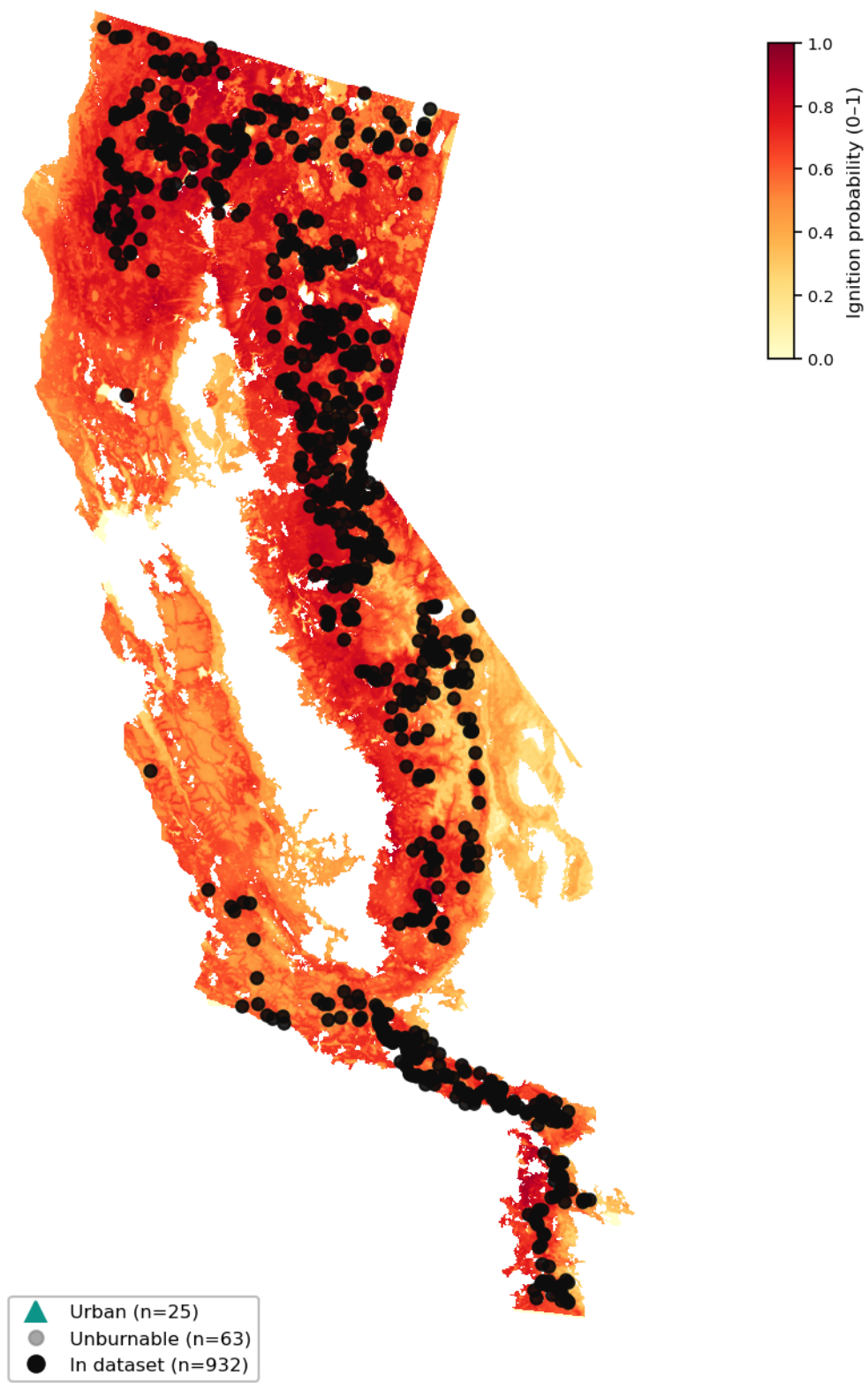


Figure 6: Dataset ignitions only

Benchmark subsample — 100 scenarios chosen with `numpy.random.default_rng(42)` without replacement from the list of scenario files that have complete config metadata, matching `run_benchmark_california2021_yearly.py` (`BENCHMARK_SUBSET_SIZE`, `RANDOM_SEED`).

5. Related documentation

- USFS pipeline and mask logic: `documentation/14_usfs_california_dataset_creation.md`
- 2021 dataset build script: `code/dataset_creation/nature_dataset_creation/create_california_2021_dataset.py`
- Mask build: `code/dataset_creation/nature_dataset_creation/build_mask_union_burnable_day1.py`

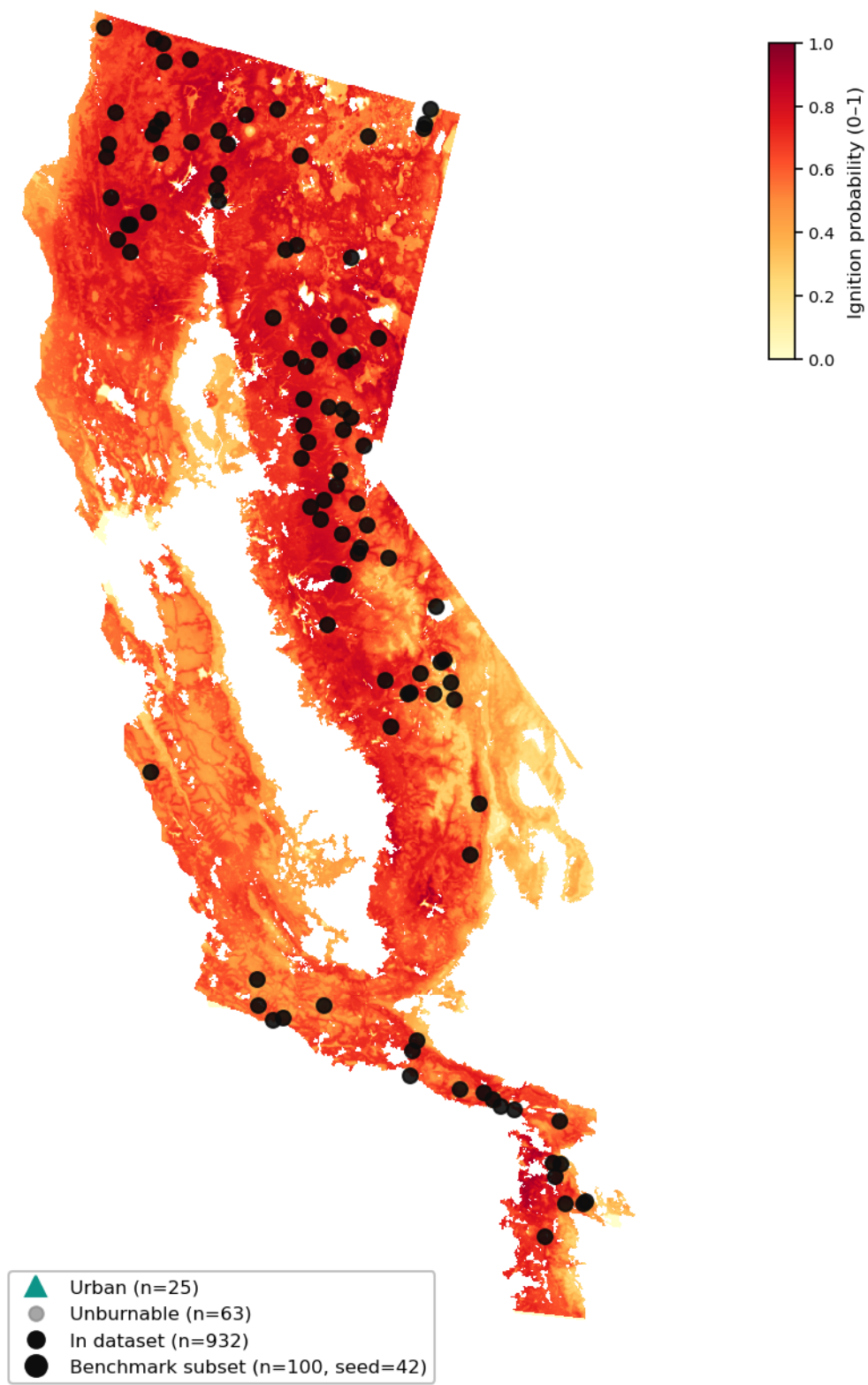


Figure 7: Benchmark 100 fires (seed 42)